



Aianox, SAU
Markijana 60
01479 – Murga (SPAIN)
www.aianox.com

INSPECTION CERTIFICATE TYPE:
CERTIFICADO DE INSPECCIÓN TIPO: EN 10204 3.1

Customer: Framo Flatøy AS

Cliente:

Dated: 28/09/2022

Fecha:

Certificate nr: 22/1454

Certificado nr:

Purchase Order nr: 1002-PO-002526

Pedido nr:

Ax internal ref: OF-22-0584

Ref. interna Ax:

Pag 1 of 2

Pág 1 de 2

Product: Casting

Producto:

Melting process: Induction furnace

Proceso de fusion:

Manufacturer brand:

Marca del fabricante:



Material specification and grade: 6A J93380 ASTM A995 & MDS
D56 rev 5 // NORSOK M-650, QTR nr 01

Material, especificación y grado:

Technical Requirements:

Requisitos Técnicos:

Pos nr.	Item No. / No. Pattern No. / Modelo	Description Descripción	Nr Pcs Nr Pzs	Heat nr Nr colada	Kg/pce Kg/pza
60000	2016-013-1	Bearing Bracket Se450	1	EZ14	354
60000	2016-013-1	Bearing Bracket Se450	1	EZ15	354
60000	2016-013-1	Bearing Bracket Se450	1	EZ16	354
60000	2016-013-1	Bearing Bracket Se450	1	EZ17	354

Chemical Analysis (% by mass) – Análisis Químico (% en peso)

Heat nr Nr colada	C	Mn	Si	P	S	Ni	Cr	Mo	Nb	Cu	N	V	W	Co	Fe
Min. Max.	0,030	1,000	1,000	0,030	0,025	6,500 8,500	24,000 26,000	3,000 4,000		0,500 1,000	0,200 0,300		0,500 1,000		
EZ14	0,024	0,750	0,650	0,023	0,001	7,720	24,770	3,890		0,610	0,280		0,650		
EZ15	0,027	0,690	0,640	0,024	0,001	7,660	24,700	3,910		0,630	0,280		0,660		
EZ16	0,026	0,710	0,760	0,022	0,001	7,580	24,790	3,770		0,590	0,260		0,610		
EZ17	0,024	0,760	0,700	0,022	0,001	7,880	24,680	3,700		0,710	0,260		0,630		



Aianox, SAU
Markijana 60
01479 – Murga (SPAIN)
www.aianox.com

INSPECTION CERTIFICATE TYPE:
CERTIFICADO DE INSPECCIÓN TIPO: EN 10204 3.1

Customer: Framo Flatøy AS		Dated: 28/09/2022		Certificate nr: 22/1454	
Cliente:		Fecha:		Certificado nr:	
Purchase Order nr: 1002-PO-002526			Ax internal ref: OF-22-0584		Pag 2 of 2
Pedido nr:			Ref. interna Ax:		Pág 2 de 2
Product: Casting		Melting process: Induction furnace		Manufacturer brand:	
Producto:		Proceso de fusion:		Marca del fabricante:	
					
Material specification and grade: 6A J93380 ASTM A995 & MDS D56 rev 5 // NORSOK M-650, QTR nr 01			Technical Requirements:		
Material, especificación y grado:			Requisitos Técnicos:		

Mechanical Properties – Propiedades Mecánicas

	Tensile Test – Ensayo de tracción						Impact test – Ensayo de resiliencia					
	ASTM A370						ASTM A370 TYPE A					
Heat nr Nr colada	Temp. °C	Rp 0,2% MPa	Rp 1% MPa	Rm MPa	A %	Z %	Temp. °C	Energy – Energía Jules			Latel exp Expn late mils	Hardness Dureza
								E1	E2	E3		
Min. Max.	20	450		700	18,0		-46	0	45	0		
EZ14	21	570		779	36,0		-46	161	162	184		
EZ15	21	529		783	39,0		-46	184	183	166		
EZ16	21	542		785	38,5		-46	150	173	149		
EZ17	21	520		774	41,0		-46	187	157	171		
Heat Treatment: Tratamiento térmico HEAT TO 1120°C,5 HOURS/ DECREASE TO 1050°C, 30`/WATER QUENCH												
Visual inspection: Inspección visual: MSS-SP-55: SATISFACTORY												
Remarks - Observaciones: PREN, Heat Nr. EZ14:42,09 - EZ15:42,08 - EZ16:41,39 - EZ17:41,05.												
Aianox certifies that the above material is in conformity with the indicated specifications, in compliance with the order. This certificate is issued electronically and is valid without signature.				Aianox certifica que el material arriba indicado cumple con las especificaciones reseñadas, según requerido en el pedido. Certificado emitido electrónicamente y válido sin firma.				Inspector: Quality Assurance: F. Oribe Aseguramiento Calidad: F.Oribe				



Aianox, SAU
Markijana 60
01479 – Murga (SPAIN)
www.aianox.com

Liquid Penetrant Examination Report

Informe de examen por líquidos penetrantes

Procedure ASME V

Report N° 22/1454

1. General Data

Customer Framo Flatøy AS

Order 1002-PO-002526

Description Bearing Bracket Se450

Pattern NR 2016-013-1

Drawing 2016-013-1

Material 6A J93380 ASTM A995 & MDS D56 rev 5

Heat NR EZ14, 1 Uds, EZ15, 1 Uds, EZ16, 1 Uds,
EZ17, 1 Uds

Surface condition As Cast

Aianox Ref.: OF-22-0584

2. Inspection Equipments

	Brand Name	Code	Colour	Batch
Penetrant	Chemetall	CHECKMORE 240	RED	
Remover		WATER		
Developer	Chemetall	LD7	WHITE	

Illumination

Natural

☐

Artificial

☒

U.V.

☐

3. Application

Surface Temperature	Penetrating time (min)	Developing time (min)
23 °C	22	22

4. Testing Place

AIANOX, S.A. (Markijana 63 01479 MURGA Alava)

5. Test Result

Casting have tested in 100% surface area by penetrant dye according ASTM E-165 and acceptance criteria according to ASME section VIII, division 1, appendix 7. Result: Satisfactory

6. Remarks

Performed and evaluated	Checked	Accepted customer
Level II EN ISO 9712 Manuel Picon Date: 18/07/2022	Level II EN ISO 9712 Manuel Picon Date: 18/07/2022	Date:
		(DAX 10010 rev.01)

**INFORME DE ENSAYO
TEST REPORT**

**Nº de Informe
Report No**
**Procedimiento - Norma
Procedure - Standard**
**Equipos
Equipment**
**Tipo de ensayo
Test type**

102802-008C-1
ASTM G-48-A (11)
53-041.02; 53-075.03; 53-037.07; 53-162.23.
PITTING CORROSION TEST

**Referencia muestras ¹⁾
Samples reference**

Order LAB 09/22 nº 08 Batches EZ14; EZ15; EZ16; EZ17.

**Dimensiones
Dimensions**

49.86 x 24.92 x 4.94 mm ; 49.79 x 24.96 x 4.97 mm.
49.82 x 24.89 x 4.96 mm ; 49.75 x 24.90 x 4.94 mm.

**Calidad ¹⁾
Quality**

Stainless steel J93380 (ASTM A995 and MDS D56 rev.5)

**Acabado superficial
Surface finishing**

120 grit + chemical pickling in (20 % HNO₃ + 5 % HF) by weight, to 60 °C for 5 minutes *

**Medio
Environment**
**Temperatura
Temperature**

FeCl₃ 6% by weight

50 °C

**Condiciones ambientales
Atmospheric conditions**
**Duración del ensayo
Test time**

-
24 h

Resultados / Results

Reference Batch	Area (cm ²)	Initial weight (g)	Final weight (g)	Weight loss		Nº of pits
				g/cm ²	g/m ²	
EZ14	32.24	47.6617	47.6617	0	0	0
EZ15	32.29	47.8721	47.8719	6.20x10 ⁻⁶	0.06	0
EZ16	32.21	47.7577	47.7577	0	0	0
EZ17	32.15	47.7860	47.7859	3.11x10 ⁻⁶	0.03	0

Observaciones / Remarks:

(*) Deviation to test standard.

**TECNALIA, realizado por
Test performed by**

J. M. González
Laboratory Technician
Date: 22/09/2022

**TECNALIA, revisado por
Reviewed by**

Andrés del Barrio
Laboratory Manager
Date: 22/09/2022



**INFORME DE ENSAYO
TEST REPORT**

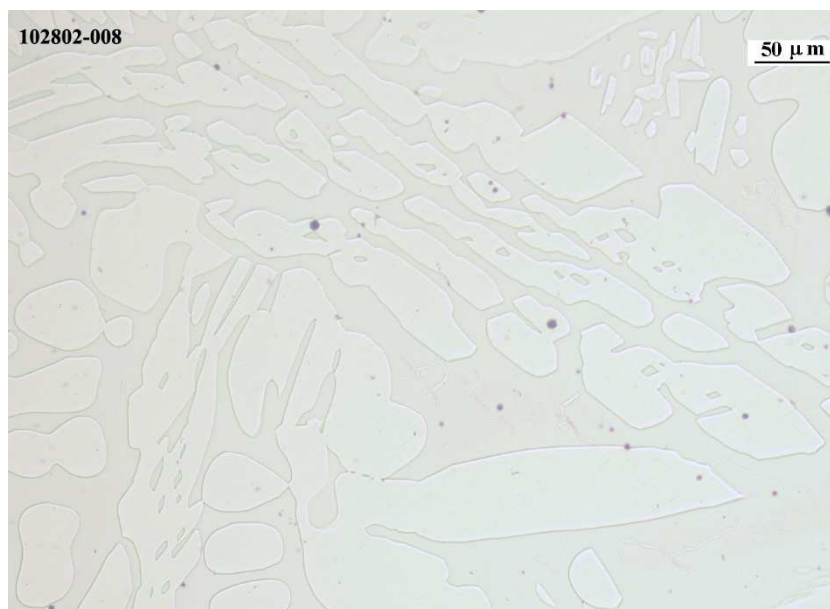
Nº Informe <i>Report No.</i>	102802-008 MT-4	Fecha de ensayo <i>Test date</i>	2022/09/19
Referencia¹⁾ <i>Reference¹⁾</i>	LAB 09/22 Nº08 Material: J93380 Coladas: EZ17		
Procedimiento - Norma <i>Procedure - Standard</i>	PE-60-MO-022 Rev.4, PE-60-MO-026 Rev.7		
Tipo de ensayo <i>Test type</i>	Determination of intermetallic phases, carbide and ferrite percentage in austenoferritic steels (metallographic method).		

Observación Micrográfica / Micrographic Observation

The micrograph (X 200) shows a detailed view of the structure which consists of austenite and ferrite.

Neither carbides nor intermetallic phases can be observed.

The ferrite mean percentage evaluated in 5 fields is: 49,1% ($\pm 2,9\%$).



Micrograph (X 200)

Observaciones/Remarks:

TECNALIA, revisado por
Reviewed by

Eneritz Zamalloa
Laboratory Manager
2022/09/19



aianox, s. a.

DIMENSIONAL CONTROL REPORT
INFORME DE CONTROL DIMENSIONAL

REPORT N°: 12863.....

CUSTOMER: FRANK MOHN FLATOY CLIENTE:		PATTERN N°: 2016-013-1 MODELO N°:		
DESCRIPTION: SOP. RODAM. SE450 DESCRIPCION:		DRAWING N°: 2016-013-1 PLANO N°:		
MATERIAL: J93380 MATERIAL:		HEAT N° : EZ17 (N°3) COLADA N°:	INSPECTION: INSPECCIONADO: Pattern /Modelo ☐ Constr. Ax ☐ Cli ☐ Piece / Pieza ☐	
INSPECTED BY: AMETS INSPECCIONADO POR:		MEASURING TABLE N° : ----- MESA DE TRAZADO N°:		
Item Pos	Measuring acc. Drawing Medida s/ plano	Control measure Medida control	Measuring instrument Instrumento de medida	Remarks Observaciones

1	Ø 820 +3/0	822.20	CD19	
2	Ø 695 0/-3	694.75	CD19	
	R.15	15	-	
5	Ø 220 0/-2	219.01	MI19	
10	80 +2/0	81.40	CD09	
12	235	234.90	CD31	

Conclusions: Conclusiones:				
ACCEPTED: ACEPTADO:	23-09-22	X	PENDING OF DECISION: PENDIENTE DE DECISION:	REJECTED: RECHAZADO:
Dimensional Control Control Dimensional				Page 1 of 1 Hoja de
A.M.S. Signature / Firma		23-09-22 Date / Fecha		

RADIOGRAPHIC REPORT

C/ P.I. Granada Fase 2 Pabellón B3
48530 - ORTUELLA (VIZCAYA)
Tlf: 946360012, Fax: 946361502
Email: ortuella@scisa.es

REPORT NBR: 220076FUNORT-OT0001-RT05
WORK 220076FUNORT-OT0001

REV/ 0



Digitally signed by:
SERVICIOS DE
CONTROL E
INSPECCIÓN S.A.
Date:2022.08.19 09:25:20

CUSTOMER: AIANOX S.A.
CUSTOMER ADDRESS: CRUCE MARKIJANA, 63, 01479, MURGA, ALAVA
TEST LOCATION: C/ P.I. GRANADA FASE 2 PABELLÓN B3, 48530, ORTUELLA, VIZCAYA

PURCHASE ORDER:	PC-22-1005	COMPONENT IDENTIFICATION:	EZ17-2016-013-1
TESTED COMPONENT:	BEARING BRACKET SE 450	MANUFACTURING STATUS:	CASTING
CUSTOMER REF.:	2016-013-1	OTHER REFERENCES:	CONTRACT: PO-002526
TEST DATE:	10/08/2022	HEAT N°:	EZ17
BATCH N°:	--	SERIAL N°:	--
EXAMINATION EXTENSION:	Z. Ind. Customer	MATERIAL:	J93380

DESCRIPTION OF EXAMINATION

PROCEDURE:	FT-18-00116	Rev:	0
TESTING STANDARD:	MDS D56 REV.5	EVALUATION STANDARD	ASME VIII DIV.1 AP.7
REFERENCE STANDARD:	ASTM E186-ASTM E186 (Steel Castings 50.8 to 114 mm in thickness - Fundición de Acero de 50.8 a 114 mm de espesor), ASTM E446-ASTM E446 (Steel Castings up to 50.8 mm in thickness - Fundición de acero hasta 50.8 mm de espesor)		
SOURCE TYPE:	Co-60	FOCAL SIZE (mm):	3.9
mA:	N/A	kV:	N/A
IQI:	Acc/Procedure	ACTIVITY (Ci):	45
MATERIAL THICKNESS:	Acc/Procedure	DENSITY:	Acc/Procedure
EXPOSITION TIME:	Various	SCREEN TYPE AND THICKNESS:	Acc/Procedure
FILM PROCESSING:	AUTOMATIC	SOURCE-FILM DISTANCE:	Acc/Procedure
FILM TYPE:	Acc/Procedure	RADIOGRAPHIC TECHNIQUE:	Simple wall
EXPOSURE TECHNIQUE SKETCH AND MARKERS:	Acc/Procedure		

N° RAD.	SECTOR	THICKNESS	DENSITY	VIEWING MODE	EVALUATION									RESULT
					POROSITY	SAND	SRINKAGE	CRACK	HOT TEAR	INSERT	MOTTILING	SURFACE	OTHER	
R1	1-2	48-86	2-4	P.D.	A1									A 
R2	2-3	48-86	2-4	P.D.										A 
R3	3-4	48-86	2-4	P.D.										A 
R4	4-5	48-86	2-4	P.D.	A1									A 
R5	5-6	48-86	2-4	P.D.	A1									A 
R6	6-7	48-86	2-4	P.D.										A 
R7	7-8	48-86	2-4	P.D.										A 
R8	8-9	48-86	2-4	P.D.										A 

RADIOGRAPHIC REPORT

REPORT NBR: 220076FUNORT-OT0001-RT05

REV/ 0

WORK 220076FUNORT-OT0001

N° RAD.	SECTOR	THICKNES S	DENSITY	VIEWING MODE	EVALUATION									
					POROSITY	SAND	SRINKAGE	CRACK	HOT TEAR	INSERT	MOTTILING	SURFACE	OTHER	RESULT
R9	9-1	48-86	2-4	P.D.										A 
R10	1-2	48-86	2-4	P.S.+P.D.			CB1							A 
R11	2-3	48-86	2-4	P.S.+P.D.			CB1							A 
R12	3-4	48-86	2-4	P.S.+P.D.			CA1							A 
R13	4-5	48-86	2-4	P.S.+P.D.										A 
R14	5-6	48-86	2-4	P.S.+P.D.										A 
R15	6-7	48-86	2-4	P.S.+P.D.			CB1							A 
R16	7-8	48-86	2-4	P.S.+P.D.		B1								A 
R17	8-9	48-86	2-4	P.S.+P.D.										A 
R18	9-10	48-86	2-4	P.S.+P.D.										A 
R19	10-1	48-86	2-4	P.S.+P.D.			CA1							A 
R20	1-2	48	2-4	P.D.										A 
R21	2-3	48	2-4	P.D.										A 
R22	3-4	48	2-4	P.D.										A 
R23	4-5	48	2-4	P.D.										A 
R24	5-6	48	2-4	P.D.										A 
R25	6-7	48	2-4	P.D.										A 
R26	7-8	48	2-4	P.D.										A 
R27	8-9	48	2-4	P.D.										A 
R28	9-1	48	2-4	P.D.										A 

GENERAL NOTES:



INTERPRETER **ROBERTO GABETE LADA**
LEVEL: II
SCI Servicios Control e Inspección. A78024668

19/08/2022

ACCEPTED CUSTOMER



KK6EI06xIB

To verify the authenticity of this certificate, access the URL
<http://www.scisa.es/verificar>

Page 2 of 2

Do not reproduce partially without written authorization from S.C.I., S.A.
Page 2 of 2 Formato F-END-3821 Rev.00

This report is only for the items listed in it.

SCI Nº: FT-18-00116 / CL

Rev. 0

Customer: AIANOX S.A.

End customer: AIANOX S.A.

Component identification : BEARING BRACKET SE 450

Material: J93380

Examination Extension: Critical zone

P/N CUSTOMER	New P/N CUSTOMER	P/N END CUSTOMER
2016-013-1 Rev.	2016-013-1 Rev.	2016-013-1 Rev.

Applicable standards

Testing standard: MDS D56 REV.5

Evaluation standard: ASME VIII DIV.1 AP.7

Equipment

Equipment	Manufacturer	Model
Cobalt	SENTINEL	680

Testing parameters

Date: 06/02/2018

Prepared by: VICTOR LALLANA ANDRES LEVEL II

Approved by: JAVIER MACIAS MALDONADO LEVEL III

SCI Nº: FT-18-00116 / CL

Rev. 0

R1;R2;R3;R4;R5				
Xray area thickness		48-86	Nbr Films	
2				
Films size Films format	Films model Films model	Films manufacturer	ASTM E1815 classification	EN-ISO 11699-1 classification
15x40	AA400	CARESTREAM	TYPE II	CLASS C5
15x40	MX125	CARESTREAM	TYPE I	CLASS C3
Front screen thickness (mm)		0.20	Front screen material	
Lead				
Back screen thickness (mm)		0.25	Back screen material	
Lead				
Intermediate screen thickness (mm)		--	Internal screen material	
--				
Front filter		--	Filter material	
--				
Backscreen		--	Backscreen material	
--				
Cassette type		Flexible	Kilovoltage	
N/A				
mA		N/A	Required density	
2-4				
Source activity		45 Co-60	Geometrical unsharpness	
0.6525292				
Source to film distance		600		
IQI		Position IQI		Essential wire/hole
35-50		Source side		2T

R6				
Xray area thickness		48-86	Nbr Films	
2				
Films size Films format	Films model Films model	Films manufacturer	ASTM E1815 classification	EN-ISO 11699-1 classification
15x40	AA400	CARESTREAM	TYPE II	CLASS C5
15x40	MX125	CARESTREAM	TYPE I	CLASS C3
Front screen thickness (mm)		0.20	Front screen material	
Lead				
Back screen thickness (mm)		0.25	Back screen material	
Lead				
Intermediate screen thickness (mm)		--	Internal screen material	
--				
Front filter		--	Filter material	
--				
Backscreen		--	Backscreen material	
--				
Cassette type		Flexible	Kilovoltage	
N/A				
mA		N/A	Required density	
2-4				
Source activity		45 Co-60	Geometrical unsharpness	
0.6525292				
Source to film distance		600		
IQI		Position IQI		Essential wire/hole
35-50		Source side		2T

SCI Nº: FT-18-00116 / CL

Rev. 0

R7;R8;R9				
Xray area thickness		48-86	Nbr Films	
2				
Films size Films format	Films model Films model	Films manufacturer	ASTM E1815 classification	EN-ISO 11699-1 classification
15x40	AA400	CARESTREAM	TYPE II	CLASS C5
15x40	MX125	CARESTREAM	TYPE I	CLASS C3
Front screen thickness (mm)		0.20	Front screen material	
Back screen thickness (mm)		0.25	Back screen material	
Intermediate screen thickness (mm)		--	Internal screen material	
Front filter		--	Filter material	
Backscreen		--	Backscreen material	
Cassette type		Flexible	Kilovoltage	
mA		N/A	Required density	
Source activity		45 Co-60	Geometrical unsharpness	
Source to film distance		600	0.6525292	
IQI		Position IQI		Essential wire/hole
35-50		Source side		2T

R10;R11;R12;R13;R14;R15;R16;R17;R18;R19				
Xray area thickness		48-86	Nbr Films	
3				
Films size Films format	Films model Films model	Films manufacturer	ASTM E1815 classification	EN-ISO 11699-1 classification
30x40	AA400	CARESTREAM	TYPE II	CLASS C5
30x40	MX125	CARESTREAM	TYPE I	CLASS C3
30x40	AA400	CARESTREAM	TYPE II	CLASS C5
Front screen thickness (mm)		0.20	Front screen material	
Back screen thickness (mm)		0.25	Back screen material	
Intermediate screen thickness (mm)		--	Internal screen material	
Front filter		--	Filter material	
Backscreen		--	Backscreen material	
Cassette type		Flexible	Kilovoltage	
mA		N/A	Required density	
Source activity		45 Co-60	Geometrical unsharpness	
Source to film distance		700	0.54625404	
IQI		Position IQI		Essential wire/hole
35-50		Source side		2T

SCI Nº: FT-18-00116 / CL

Rev. 0

R20;R21;R22;R23;R24;R25;R26;R27;R28

Xray area thickness		48	Nbr Films		2
Films size Films format	Films model Films model	Films manufacturer		ASTM E1815 classification	EN-ISO 11699-1 classification
30x40	T200	CARESTREAM		TYPE I	CLASS C4
30x40	T200	CARESTREAM		TYPE I	CLASS C4
Front screen thickness (mm)		0.20	Front screen material		Lead
Back screen thickness (mm)		0.25	Back screen material		Lead
Intermediate screen thickness (mm)		--	Internal screen material		--
Front filter		--	Filter material		--
Backscreen		--	Backscreen material		--
Cassette type		Flexible	Kilovoltage		N/A
mA		N/A	Required density		2-4
Source activity		10 Co-60	Geometrical unsharpness		0.2871166
Source to film distance		700			
IQI		Position IQI		Essential wire/hole	
35		Source side		2T	

Radiographic sketch

2016-013-1

Associated views/Associated views: R1 R2 R3 R4 R5 R6 R7 R8 R9 R20 R28



Radiographic sketch

2016-013-1

Associated views/Associated views: R2 R3 R4 R5 R21 R22 R23



2016-013-1

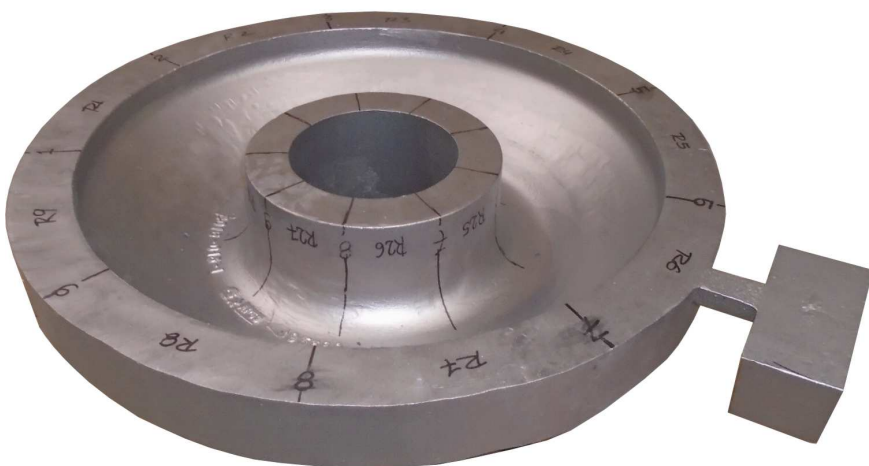
Associated views/Associated views: R4 R5 R6 R7 R23 R24 R25



Radiographic sketch

2016-013-1

Associated views/Associated views: R6 R7 R8 R9 R25 R26 R27



2016-013-1

Associated views/Associated views: R1 R10 R19 R20 R27 R28



Radiographic sketch

2016-013-1

Associated views/Associated views: R11 R12 R13 R21 R22 R23



2016-013-1

Associated views/Associated views: R14 R15 R16 R23 R24 R25



Radiographic sketch

2016-013-1

Associated views/Associated views: R17 R18 R19 R26 R27



NORMAS DE EVALUACIÓN PARA ACEROS MOLDEADOS
EVALUATION STANDARDS FOR STEEL CASTINGS

ASTM- E71 Radiografías de referencia para aceros moldeados de hasta 2 pulgadas de (51 mm) de espesor <i>Reference radiographs for steel castings Up to 2 in (52 mm) thickness</i>		ASTM- E446 Radiografías de referencia para aceros moldeados de hasta 2 pulgadas de (51 mm) de espesor <i>Reference radiographs for steel castings Up to 2 in (52 mm) thickness</i>	
Tipo de discontinuidad <i>Discontinuity type</i>	Designación y niveles de severidad <i>Designation & Severity levels</i>	Tipo de discontinuidad <i>Discontinuity type</i>	Designación y niveles de severidad <i>Designation & Severity levels</i>
Porosidad <i>Gas & Blowholes</i>	A- (1-6)	Porosidad <i>Gas porosity</i>	A – (1-5)
Arenas e inclusiones <i>Sand spots & inclusions</i>	B- (1-6)	Inclusiones de arena y escorias <i>Sand & slag inclusions</i>	B – (1-5)
Rechue interno <i>Internal shrinkage</i>	C- (1-6)	Rechue tipo 1 <i>Shrinkage type 1</i>	CA – (1-5)
Desgarros en caliente <i>Hot Tears</i>	D- (1-3)	Rechue tipo 2 <i>Shrinkage type 2</i>	CB – (1-5)
Grietas <i>Cracks</i>	E- (1-3)	Rechue tipo 3 <i>Shrinkage type 3</i>	CC – (1-5)
Soportes de macho no fundido <i>Unfused chaplets</i>	F- (1-3)	Rechue tipo 4 <i>Shrinkage type 4</i>	CD – (1-5)
Enfriadores internos (no fundidos) <i>Internal Chills</i>	G- (1-4)	Grietas <i>Crack</i>	D –one illustration
		Desgarro en Caliente <i>Hot Lear</i>	E- one illustration
		Inserción <i>(Insert)</i>	F- one illustration
		Moteado <i>Mottling</i>	G- one illustration
ASTM- E186 Radiografías de referencia para aceros moldeados De paredes gruesas [2 a 4 ½ pulgadas (51 – 114 mm)] <i>Reference radiographs for heavy-walled [2 to 4 ½ in. (51 to 114 mm)]</i>		ASTM- E280 Radiografías de referencia para aceros moldeados De paredes gruesas [2 a 4 ½ pulgadas (114 – 305 mm)] <i>Reference radiographs for heavy-walled [2 to 4 ½ in. (114 to 305 mm)]</i>	
Tipo de discontinuidad <i>Discontinuity type</i>	Designación y niveles de severidad <i>Designation & Severity levels</i>	Tipo de discontinuidad <i>Discontinuity type</i>	Designación y niveles de severidad <i>Designation & Severity levels</i>
Porosidad <i>Gas porosity</i>	A – (1-5)	Porosidad <i>Gas porosity</i>	A – (1-5)
Inclusiones de arena y escorias <i>Sand & slag inclusions</i>	B – (1-5)	Inclusiones de arena y escorias <i>Sand & slag inclusions</i>	B – (1-5)
Rechue tipo 1 <i>Shrinkage type 1</i>	CA – (1-5)	Rechue tipo 1 <i>Shrinkage type 1</i>	CA – (1-5)
Rechue tipo 2 <i>Shrinkage type 2</i>	CB – (1-5)	Rechue tipo 2 <i>Shrinkage type 2</i>	CB – (1-5)
Rechue tipo 3 <i>Shrinkage type 3</i>	CC – (1-5)	Rechue tipo 3 <i>Shrinkage type 3</i>	CC – (1-5)
Grietas <i>Crack</i>	D –one illustration	Grietas <i>Crack</i>	D –one illustration
Desgarro en Caliente <i>Hot Lear</i>	E- one illustration	Desgarro en Caliente <i>Hot Lear</i>	E- one illustration
Inserción <i>(Insert)</i>	F- one illustration	Inserción <i>(Insert)</i>	F- one illustration

NORMAS DE EVALUACIÓN PARA FUNDICIONES GRISES
EVALUATION STANDARDS FOR GRAY IRON CASTINGS

A ASTM E 446 ó E188, añadir ASTM – E 802 (Rechue en pluma - niveles 1-5)
 To ASTM – E 446 or E 186, add ASTM – E 802 (Feathery Shrinkage - Levels 1-5)

ABREVIATURAS SYMBOLS	Modo de Observación <i>Viewing mode</i>	Felicula Sencilla / <i>Single film</i> Película Doble / <i>Composite film</i>	“P.S.” “P.D.”
	Ningún Defecto <i>No Defects</i>	S.D.A. (Sin Defectos aparentes)	
	Defectos superficiales	Ligero / <i>Small</i> Medio / <i>Medium</i> Grueso / <i>Large</i>	“L” “M” “G”
	Zona Radiografiada <i>Radiographic area</i>	Aceptada / <i>Accepted</i> Rechazada / <i>Rejected</i>	“A” “R”